

Sleep Data Analysis: Exploring the Science of Rest

This analytical presentation details findings from a comprehensive Sleep Data Analysis project performed using Python and key data science libraries. Our goal is to uncover significant relationships between lifestyle factors and sleep quality, providing actionable insights for improving well-being and cognitive performance.

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Professor: Dr.Tahmores

Project Foundation and Dataset Overview

Research Context

This project examines health sleep statistics to identify patterns and correlations between daily habits and rest quality. Using rigorous statistical methods and machine learning techniques, we transform raw data into meaningful insights applicable to student life and beyond.

Target Audience: Computer Science Students

Data Source: Health Sleep Statistics Dataset

Analytical Approach: Scientific, evidence-based

Dataset Specifications

- **Total Records:** 100 participants
- **Features:** 12 distinct variables
- **Data Quality:** Zero missing values
- **Data Types:** Mixed numerical and categorical
- **Integrity Check:** No duplicate entries detected

The pristine quality of this dataset enables direct modeling and analysis without extensive preprocessing, allowing us to focus on extracting meaningful patterns and relationships.

Data Characteristics and Key Variables

Core Metrics Analyzed

- **Age:** 22-50 years (mean: 36)
- **Sleep Quality:** 4-9 scale (mean: 7.0)
- **Daily Steps:** 3,000-11,000 range
- **Calories Burned:** 2,000-2,900 range
- **Sleep Duration:** 5.5-8.0 hours

Summary statistics revealed healthy ranges across all numerical variables, confirming the dataset's suitability for predictive modeling and correlation analysis.

Categorical Features

The dataset includes several categorical variables that provide context for behavioral patterns:

- **Gender:** Male and female participants
- **Physical Activity Level:** Low, medium, high (coded 1-3)
- **Dietary Habits:** Unhealthy, medium, healthy (coded 1-3)
- **Sleep Disorders:** Presence or absence
- **Medication Usage:** Yes or no

These categorical variables were strategically encoded into numerical values to enable quantitative analysis while preserving their ordinal relationships and semantic meaning.

Technical Environment and Analytical Tools



Data Handling

NumPy & Pandas: Essential for array manipulation and structuring raw CSV data into manageable DataFrames for efficient querying and transformation.



Visualization

Matplotlib & Seaborn: Generate descriptive charts including bar plots, histograms, and heatmaps to visually inspect data distributions and identify relationships.



Modeling & Statistics

Scikit-Learn & SciPy: Employed for predictive linear regression modeling and rigorous statistical hypothesis testing using T-tests and correlation analysis.

The Python ecosystem provides a comprehensive toolkit for every stage of the data science pipeline, from initial exploration through statistical validation to final visualization. This standardized approach ensures reproducibility and scientific rigor throughout our analysis.

Dataset Overview

```
print("=== Dataset Shape ===")
print(df.shape)

print("\n=== Basic Info ===")
print(df.info())
```

```
=== Dataset Shape ===
(100, 12)
```

```
=== Basic Info ===
<class 'pandas.core.frame.DataFrame’>
RangeIndex: 100 entries, 0 to 99Data columns
(total 12 columns):
 #   Column                    Non-Null Count
Dtype ---  ------                    --------------
-----  0   User ID                   100 non-null
int64  1   Age                       100 non-null
int64  2   Gender                    100 non-null
object 3   Sleep Quality             100 non-null
int64  4   Bedtime                   100 non-null
object 5   Wake-up Time              100 non-null
object 6   Daily Steps               100 non-null
int64  7   Calories Burned           100 non-null
int64  8   Physical Activity Level   100 non-null
object 9   Dietary Habits            100 non-null
object 10  Sleep Disorders           100 non-null
object 11  Medication Usage          100 non-null
objectdtypes: int64(5), object(7)memory usage:
9.5+ KBNone
```

Dataset Overview

```
print("=== Summary Statistics ===")
print(df.describe())
```

```
=== Summary Statistics ===
          User ID         Age  Sleep Quality   Daily Steps  Calories Burned
count  100.000000  100.000000     100.000000    100.000000       100.00000
mean    50.500000   36.010000       7.000000   6830.000000      2421.00000
std     29.011492    8.454865       1.734964   2498.706736       281.06759
min      1.000000   22.000000       4.000000   3000.000000      2000.00000
25%     25.750000   28.750000       5.750000   4750.000000      2175.00000
50%     50.500000   35.000000       7.500000   6750.000000      2400.00000
75%     75.250000   44.000000       8.250000   9000.000000      2700.00000
max    100.000000   50.000000       9.000000  11000.000000      2900.00000
```

Dataset Overview

```
print("=== Missing Values ===")
print(df.isna().sum())
```

```
=== Missing Values ===
User ID                          0
Age                              0
Gender                           0
Sleep Quality                    0
Bedtime                          0
Wake-up Time                     0
Daily Steps                      0
Calories Burned                  0
Physical Activity Level          0
Dietary Habits                   0
Sleep Disorders                  0
Medication Usage                 0
dtype: int64
```

Dataset Overview

```
print("\n=== Duplicate Rows ===")
print(df.duplicated().sum())
print("\n=== Columns ===")
print(df.columns)
```

```
=== Duplicate Rows ===
0

=== Columns ===
Index(['User ID', 'Age', 'Gender', 'Sleep Quality', 'Bedtime', 'Wake-up Time',
       'Daily Steps', 'Calories Burned', 'Physical Activity Level',
       'Dietary Habits', 'Sleep Disorders', 'Medication Usage'],
      dtype='object')
```

Age vs Sleep Quality

```
age_vs_sleep_quality_corr = df[['Age', 'Sleep Quality']].corr()
print("\n=== Correlation: Age vs Sleep Quality ===")
print(age_vs_sleep_quality_corr)
```

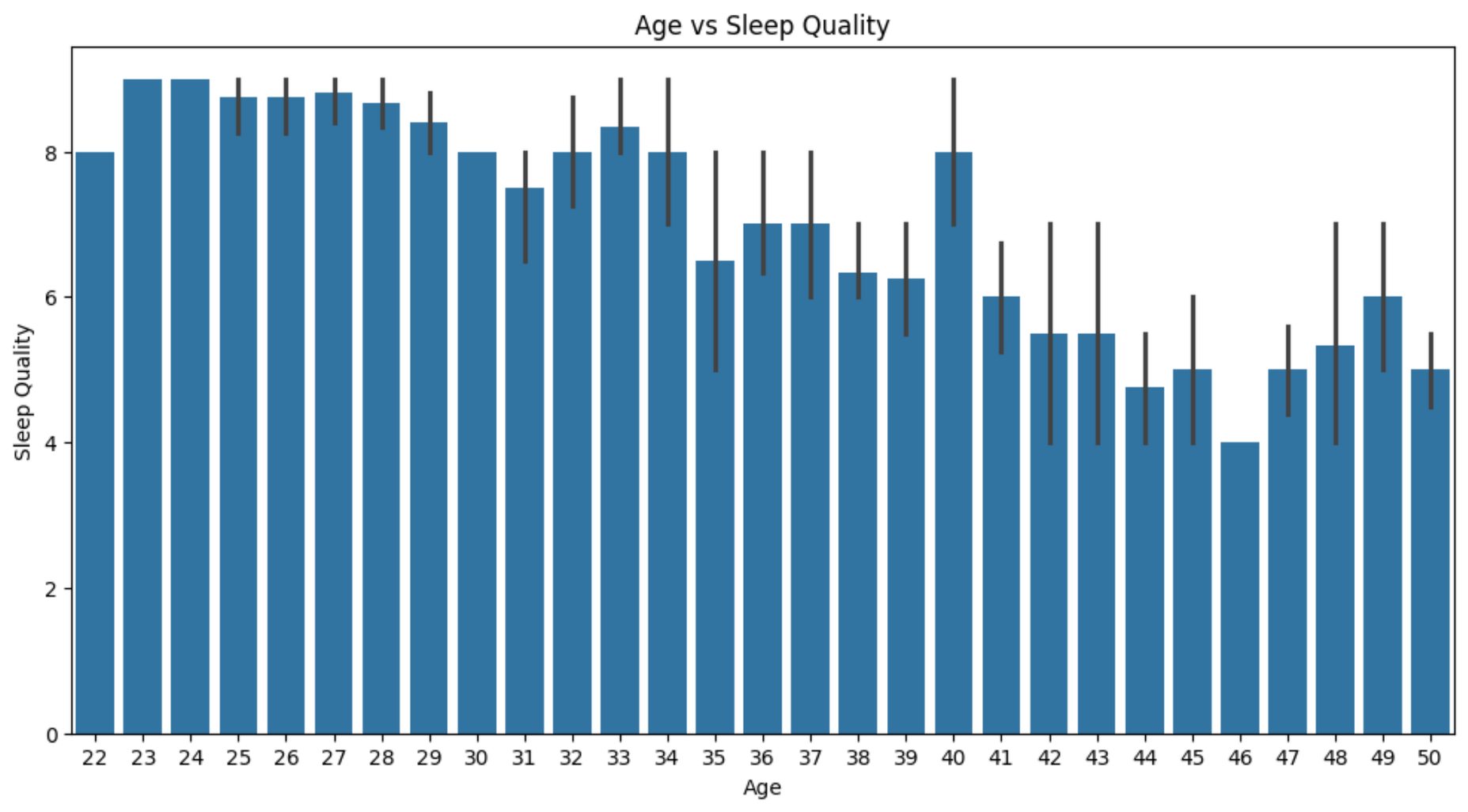
```
=== Correlation: Age vs Sleep Quality ===
                    Age  Sleep Quality
Age            1.000000      -0.848356
Sleep Quality -0.848356       1.000000
```

The `.corr()` function in Python (using pandas) is used to calculate the **Pearson correlation coefficient** between numerical variables.
It measures how much two variables change together (covariance), normalized by their individual variability (standard deviation).

The value **1.000000** on the main diagonal of the matrix indicates a **perfect correlation** of each variable with itself.
The value **-0.848356** shows a **very strong negative correlation** between **Age** and **Sleep Quality**.
This means that younger individuals generally tend to have higher-quality sleep, while sleep quality significantly declines with increasing age.



Age vs Sleep Quality





Age vs Sleep Quality

```
X = df[['Age']]
Y = df['Sleep Quality']
model = LinearRegression()
model.fit(X, Y)
```

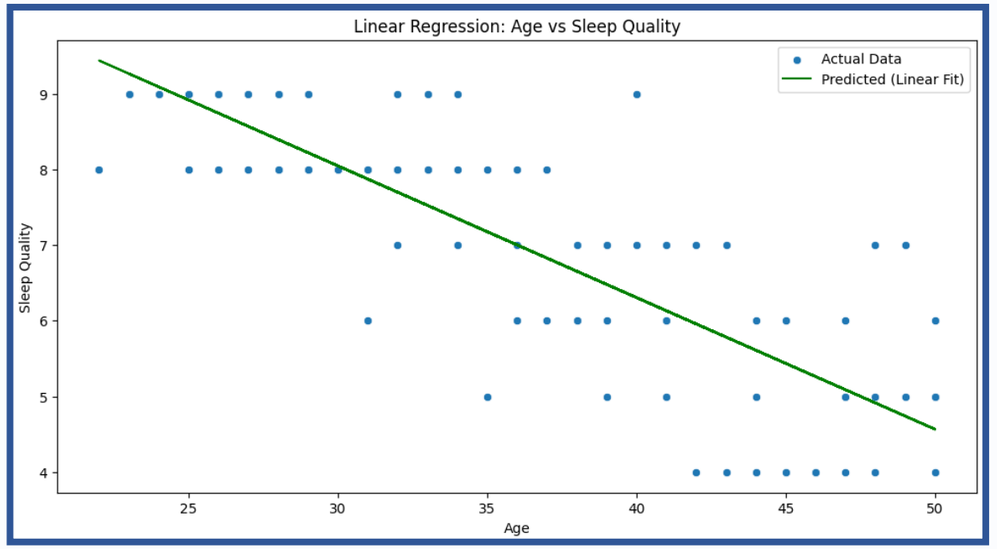
```
plt.figure(figsize=(12,6))
sns.scatterplot(x='Age', y='Sleep Quality', data=df, label='Actual Data')
plt.plot(X, model.predict(X), color='green', label='Predicted (Linear Fit)')
plt.title("Linear Regression: Age vs Sleep Quality")
plt.legend()
plt.show()

print("Model Coefficient:", model.coef_)
print("Model Intercept:", model.intercept_)
```



Age vs Sleep Quality

```
Model Coefficient: [-0.17408531]
Model Intercept: 13.268812023190646
```



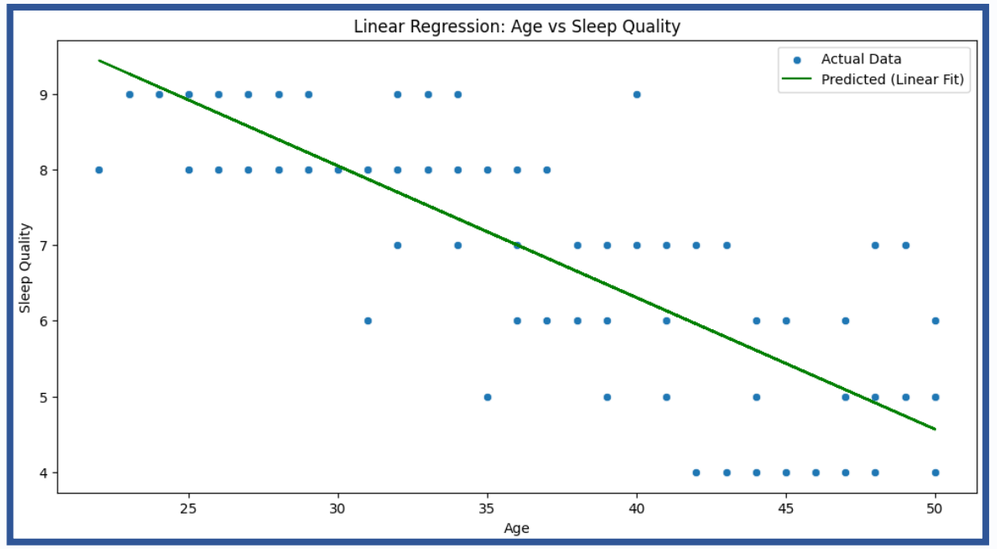
Model Results:
Coefficient: -0.174
➜ For every one-year increase in age, sleep quality decreases by an average of **0.174 units**.
Intercept: 13.27
➜ Represents the predicted sleep quality for an individual with age zero (the model's baseline).

➢ The **negative trend** indicates that as age increases, **sleep quality tends to decrease**



Age vs Sleep Quality

```
Model Coefficient: [-0.17408531]
Model Intercept: 13.268812023190646
```



Model Results:
Coefficient: -0.174
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T-test Concept

The **T-test** is a statistical method used to **compare the means of two groups** to determine whether the observed difference is **real and meaningful**, or merely **due to random variation** in the data.

T-Statistic: Indicates the degree of difference between the two group means.

Larger absolute values (positive or negative) → greater difference between groups.

A **negative sign** means the first group (males) has a lower mean than the second group (females).

P-Value: Represents the probability that the observed difference occurred by chance.

If **P < 0.05** → the difference is **statistically significant** (likely real).

If **P > 0.05** → the difference is **not significant** (possibly random).

Result in Our Data:

Since the P-value for both tests was approximately **0.0000**,
it means that the differences between males and females in **sleep quality** and **physical activity** are **real and statistically significant.**





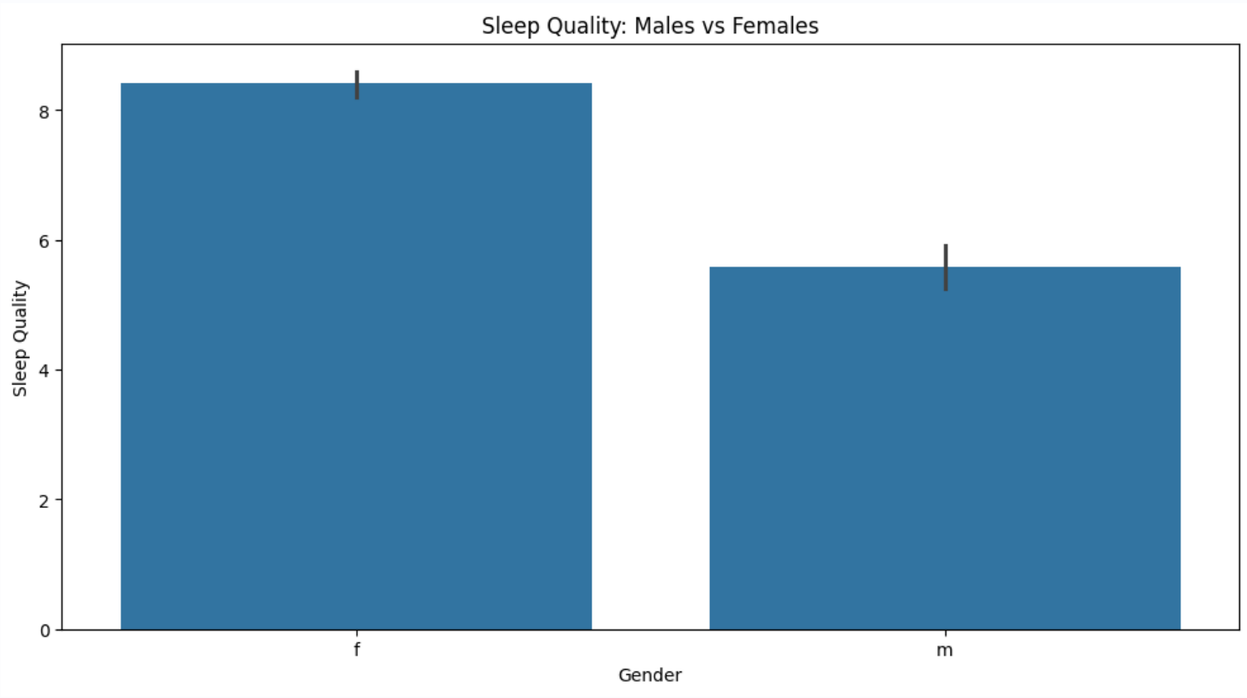
Sleep Quality: Males vs. Females

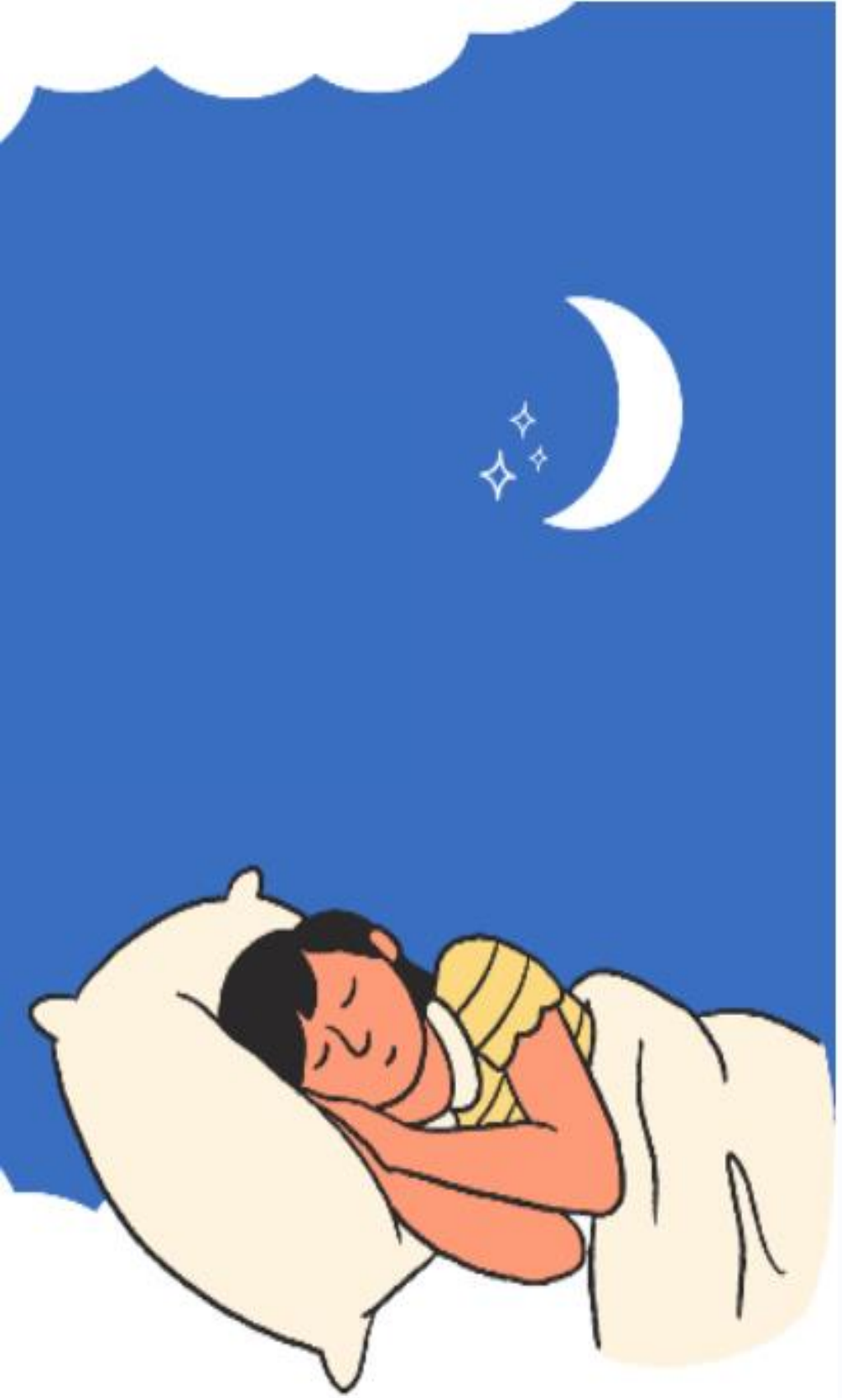
- This chart shows the **average sleep quality** for males and females.
- The data reveal that **females tend to have higher sleep quality** than males.
- A **T-test** was performed to statistically examine this difference.

T-test Results:
- **T = -14.32**
- **P = 0.0000**

→ The P-value is extremely small, meaning the observed difference is **statistically significant and not random**.

→ **Conclusion:** There is a **significant statistical difference** in sleep quality between males and females





Calculating Users' Sleep Duration

•Convert Bedtime and Wake-up Time data into numerical values to calculate sleep duration in hours for analysis.

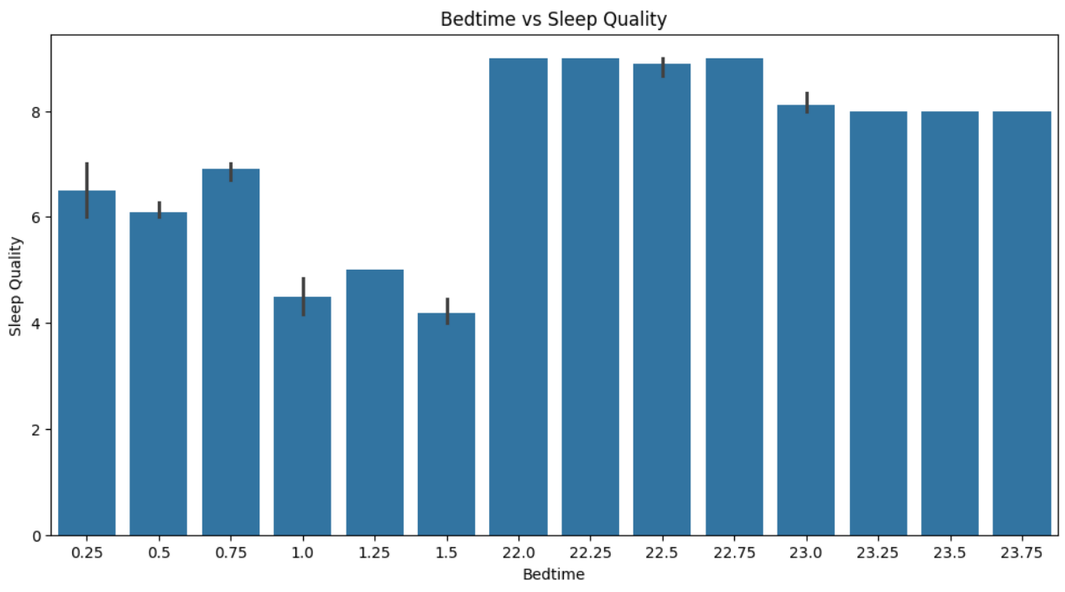
```
def sleep_duration_hours_helper (str_time):
    h = int(str_time.split(':')[0])
    m = int(str_time.split(':')[1])
    return h + m/60

df['Bedtime’] = df['Bedtime'].apply(sleep_duration_hours_helper)
df['Wake-up Time'] = df['Wake-up
Time'].apply(sleep_duration_hours_helper)

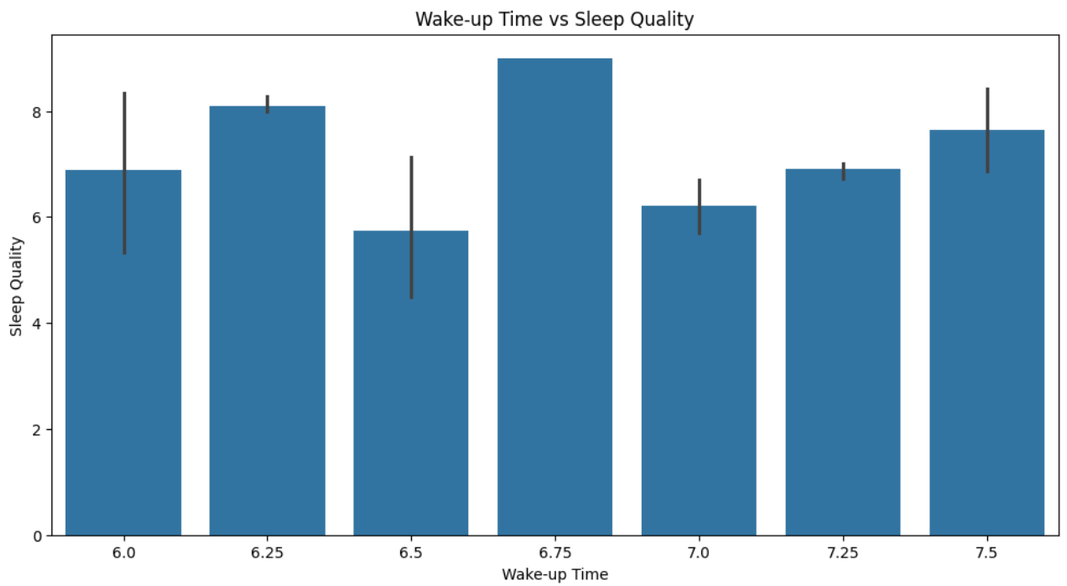
df['Sleep Duration Hours'] = (df['Wake-up Time'] -
df['Bedtime']) % 24
print(df[['Bedtime', 'Wake-up Time', 'Sleep Duration
Hours']].head())
df.head()
```

	Bedtime	Wake-up Time	Sleep Duration Hours
0	23.00	6.50	7.5
1	0.50	7.00	6.5
2	22.75	6.75	8.0
3	1.00	6.50	5.5
4	23.50	7.00	7.5

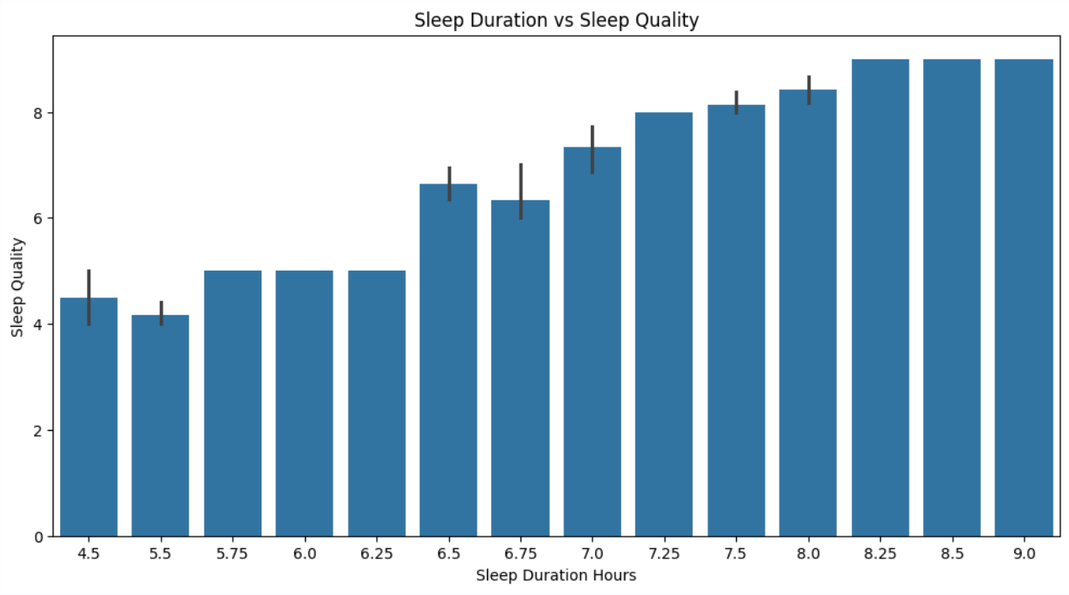
Bedtime vs Sleep Quality



Wake-up Time vs Sleep Quality



Sleep Duration vs Sleep Quality



Comparing Average Sleep and Activity Metrics by Gender

To examine the differences in average health and sleep-related variables between males and females.

```
gender_summary = df.groupby('Gender')[['Sleep Quality', 'Physical Activity Level',
'Calories Burned', 'Sleep Duration Hours']].mean().round(2)
print("\n=== Average Metrics by Gender ===")
print(gender_summary)
```

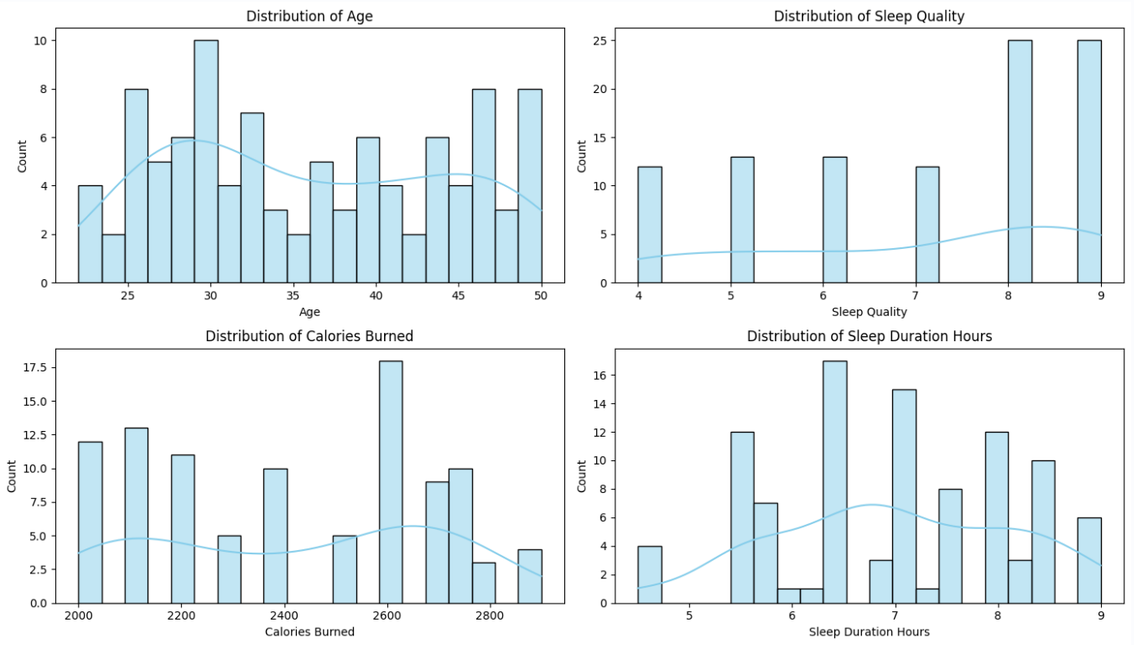
Gender	Sleep Quality	Physical Activity Level	Calories Burned	Sleep Duration Hours
Female (f)	8.42	2.68	2654.0	7.89
Male (m)	5.58	1.52	2188.0	6.10

- On average, **females have higher sleep quality** and **longer sleep duration** than males.
- Females also show **higher physical activity levels** and **greater calorie expenditure**

Comparing Average Sleep and Activity Metrics by Gender

To explore the overall patterns in key numerical variables such as **Age**, **Sleep Quality**, **Calories Burned**, and **Sleep Duration**. By visualizing the distribution of each feature, we can understand where the data are concentrated and identify underlying trends or groups.

```
numeric_cols = ['Age', 'Sleep Quality',
'Calories Burned', 'Sleep Duration
Hours']
plt.figure(figsize=(14,8))
for i, col in enumerate(numeric_cols,
1):
    plt.subplot(2, 2, i)
    sns.histplot(df[col], kde=True,
bins=20, color='skyblue')
    plt.title(f"Distribution of {col}")
plt.tight_layout()
plt.show()
```



- These distribution charts provide valuable insight into the **structure and variability** of the dataset, helping us better understand the data **before performing statistical tests or modeling.**

Sleep Duration Analysis: Quantity Predicts Quality

Methodology

- Sleep duration was calculated as the difference between **Bedtime** and **Wake-up Time** (considering the 24-hour cycle).
  The **average sleep duration** of participants ranged from **6 to 8 hours**.
- Participants who **go to bed earlier** and **sleep longer** consistently exhibit **higher sleep quality**; this pattern is observed across all age and demographi groups.

Key Finding: Earlier bedtimes combined with longer total sleep duration consistently yielded higher sleep quality scores across all age groups and demographics.

Regression Model Performance

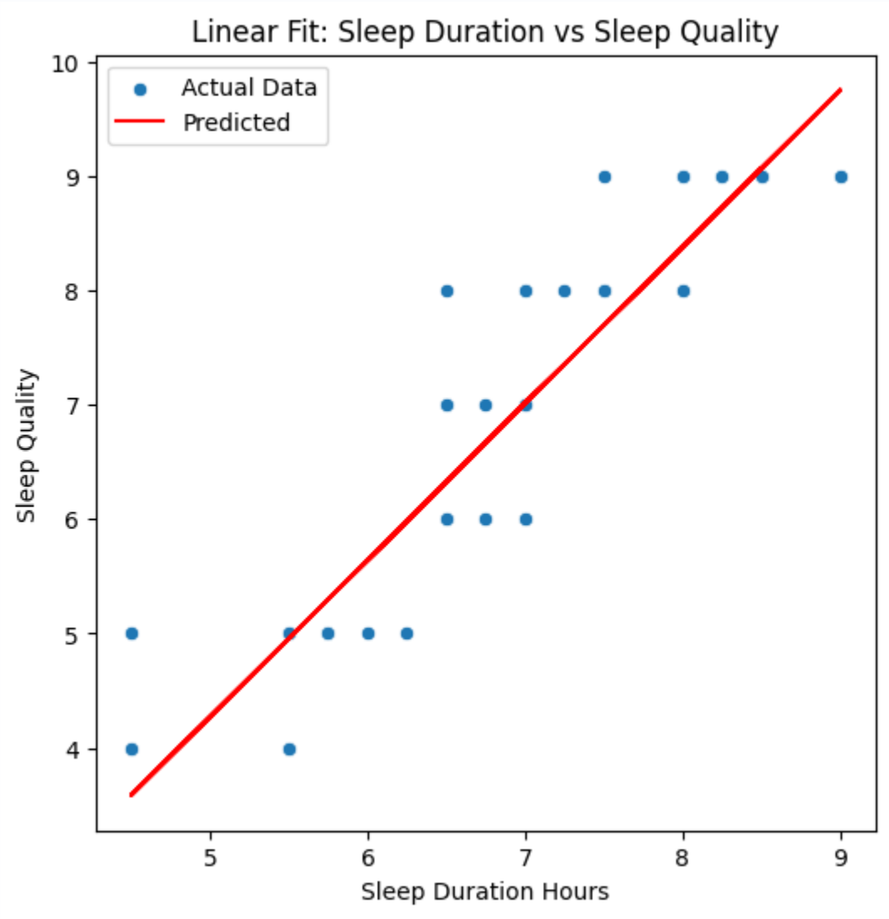
0.84

R^2 Score

This high value demonstrates a **strong positive relationship** between sleep duration and sleep quality.

Sleep duration is one of the **strongest predictors of sleep quality**. Getting sufficient sleep (around **7 to 8 hours**) can be the **most effective action** to improve sleep quality.

Sleep Duration Analysis: Quantity Predicts Quality



Regression Model Performance

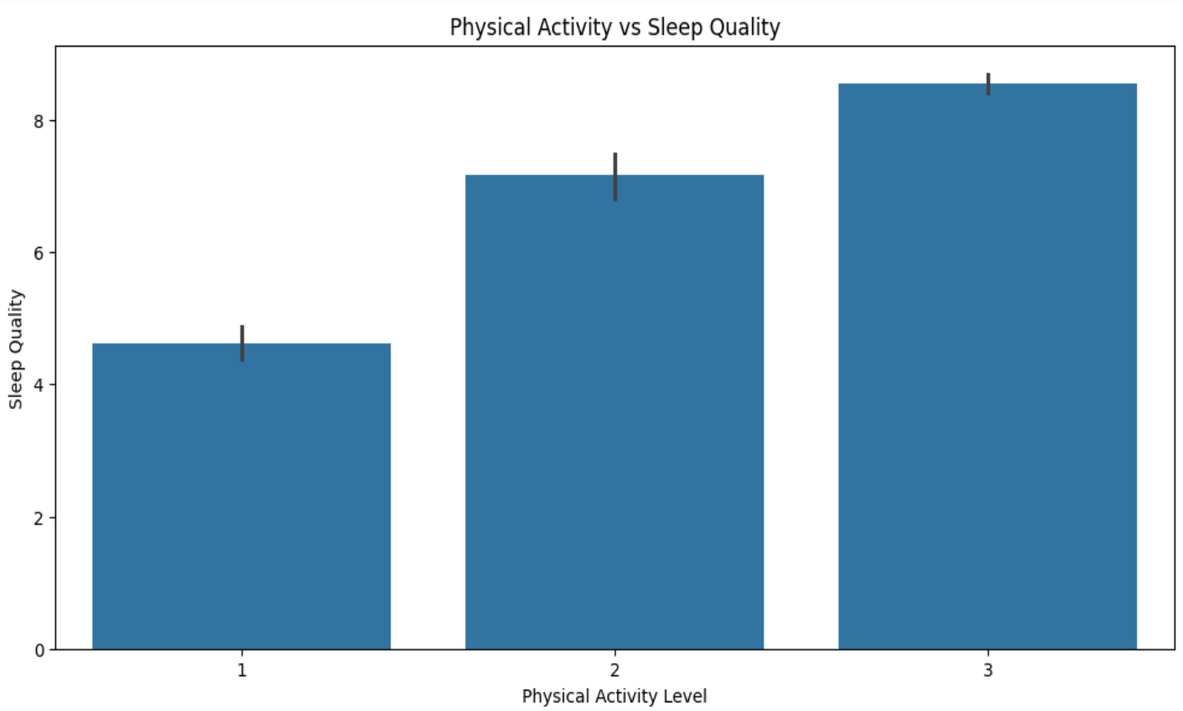
0.84

R^2 Score

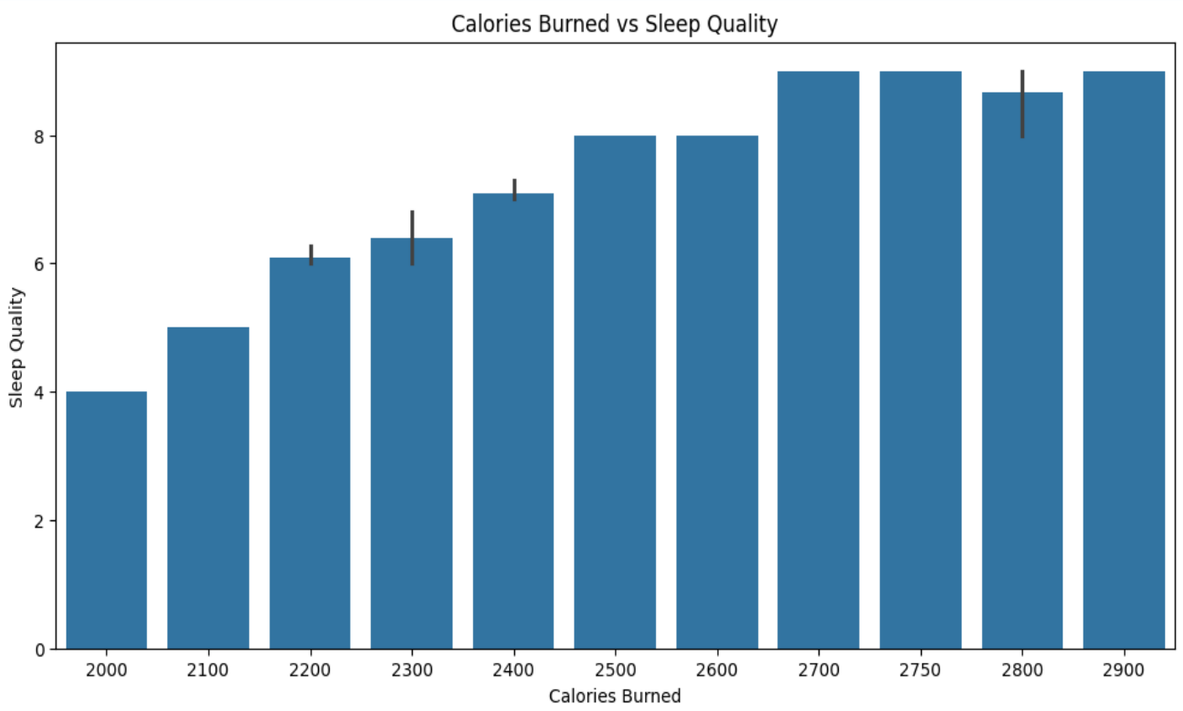
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Physical Activity vs Sleep Quality



Calories Burned vs Sleep Quality



Sleep Duration Analysis: Quantity Predicts Quality

Methodology

- To investigate whether the amount of **calories burned** during the day can predict **sleep quality**.

- A Linear Regression model was used to examine the relationship between calories burned and sleep quality.
- Data points represent actual values, while the red line shows the model's predictions.

Key Finding: Earlier bedtimes combined with longer total sleep duration consistently yielded higher sleep quality scores across all age groups and demographics.

Regression Model Performance

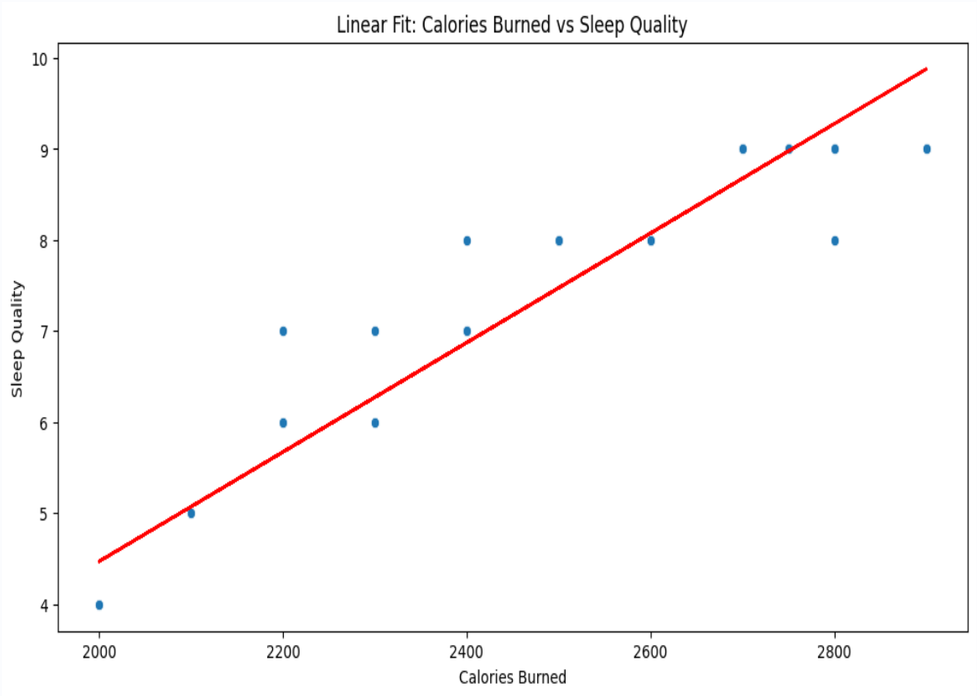
0.94

R^2 Score

This indicates that **daily physical activity and calorie expenditure** are among the **strongest predictors of sleep quality** in this dataset.

Engaging in sufficient physical activity and burning calories during the day is **directly associated with better sleep quality** and can be one of the **most effective factors for improving sleep**.

Sleep Duration Analysis: Quantity Predicts Quality



Regression Model Performance

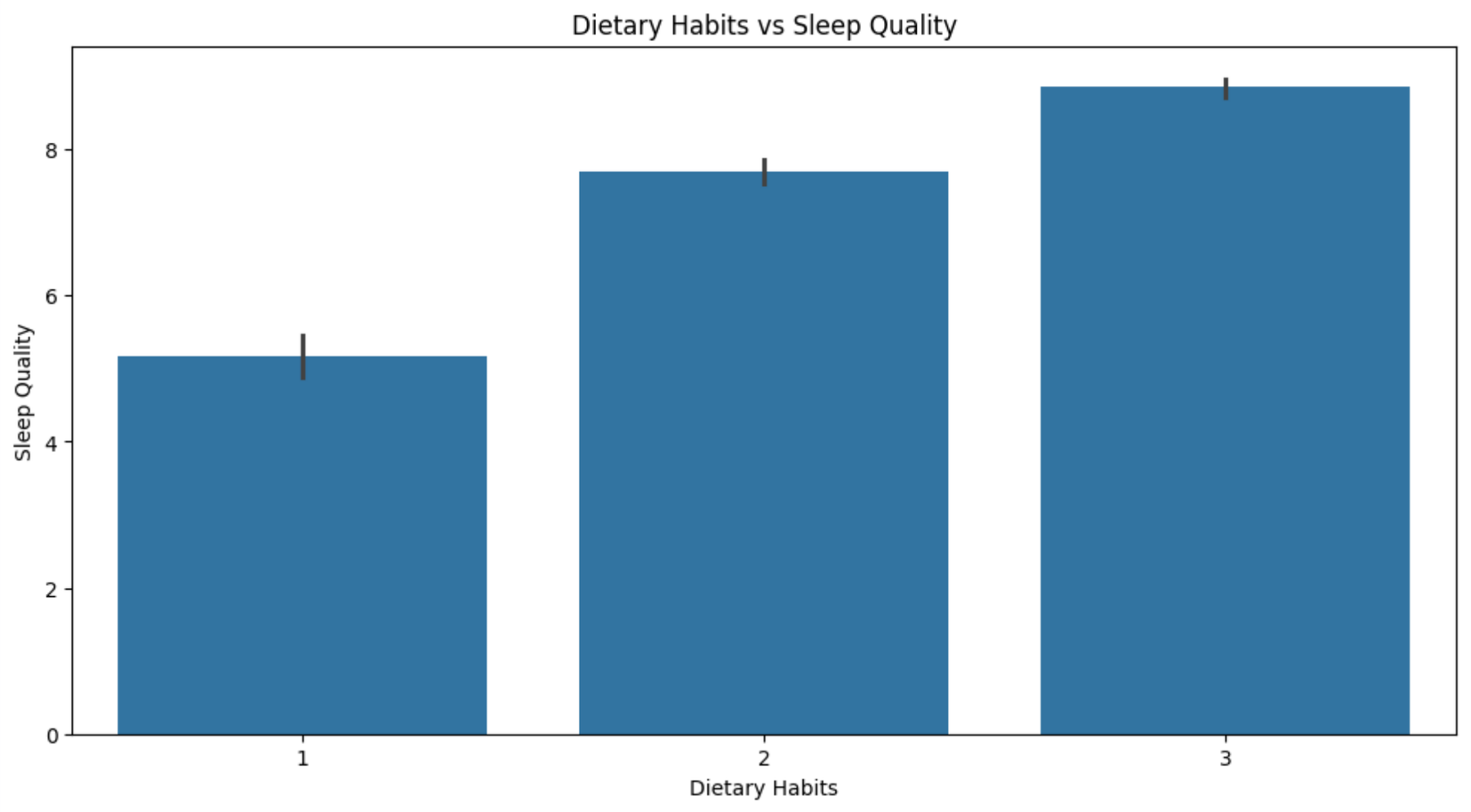
0.94

R^2 Score

This indicates that **daily physical activity and calorie expenditure** are among the **strongest predictors of sleep quality** in this dataset.

Engaging in sufficient physical activity and burning calories during the day is **directly associated with better sleep quality** and can be one of the **most effective factors for improving sleep**.

Dietary Habits vs Sleep Quality



Impact of Sleep Disorder on Sleep Quality

To examine how having a **sleep disorder** affects **sleep quality** and **physical activity levels**.

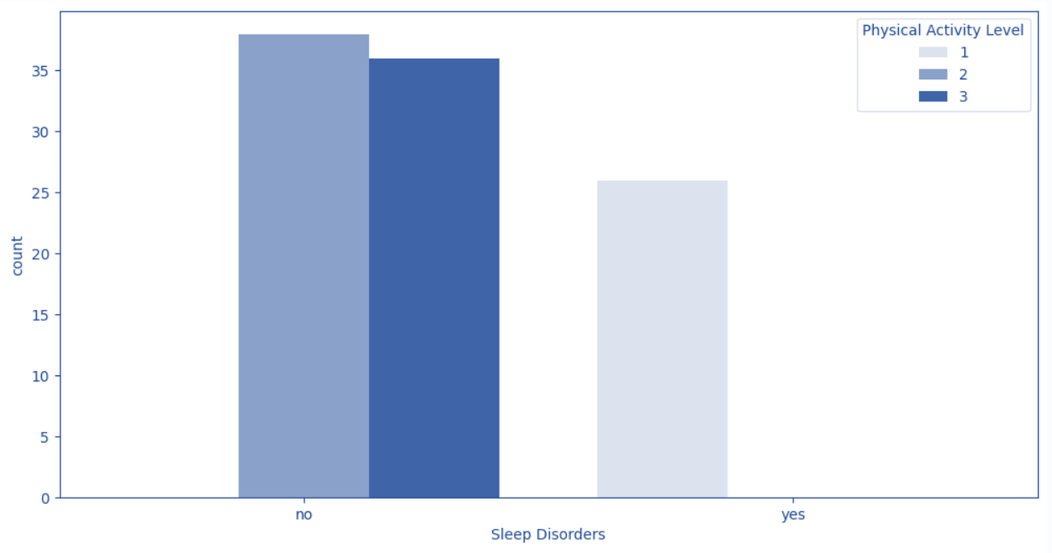
- Compared **mean and standard deviation** of sleep quality between groups with and without sleep disorders.
- Examined the **distribution of physical activity levels** in each group.
- Observed differences using **box plots** and **count plots**.

Statistical Results:

•**T-test** shows a highly significant difference in sleep quality between the two groups.

•**t = -14.12**, **p = 0.00** → the impact of sleep disorder on sleep quality is **statistically significant**.

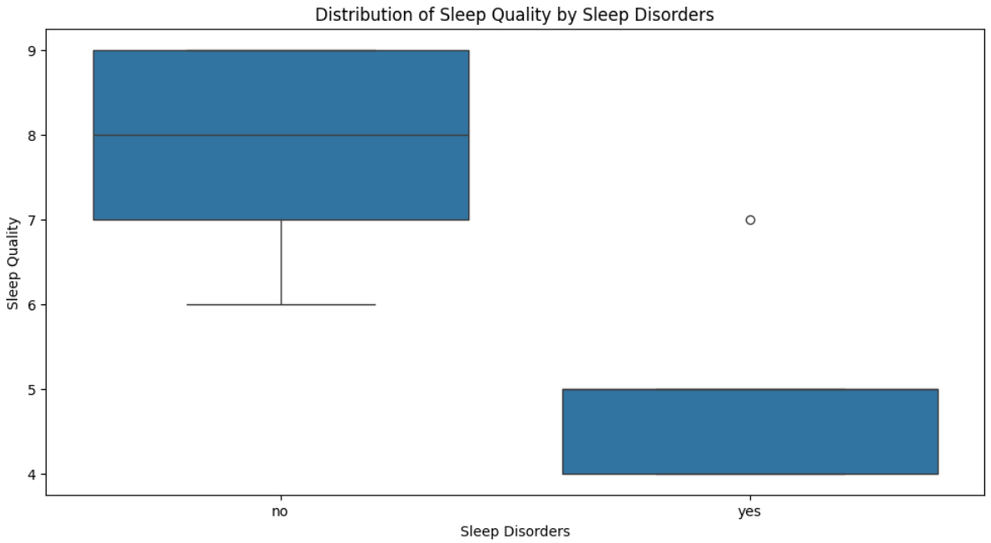
Individuals with **sleep disorders** have **lower sleep quality**.
Their **physical activity distribution** differs markedly from those without sleep disorders.



Sleep Disorder Group	Mean Sleep Quality	Std. Deviation
No (without)	7.84	1.09
Yes (with)	4.62	0.70

Sleep Quality Distribution by Sleep Disorder

Sleep Disorder Group	Mean Sleep Quality	Std. Deviation
No (without)	7.84	1.09
Yes (with)	4.62	0.70



Having a **sleep disorder significantly reduces sleep quality** and is associated with **lower physical activity levels**.

Medication Usage vs Sleep Quality

Statistical Results:

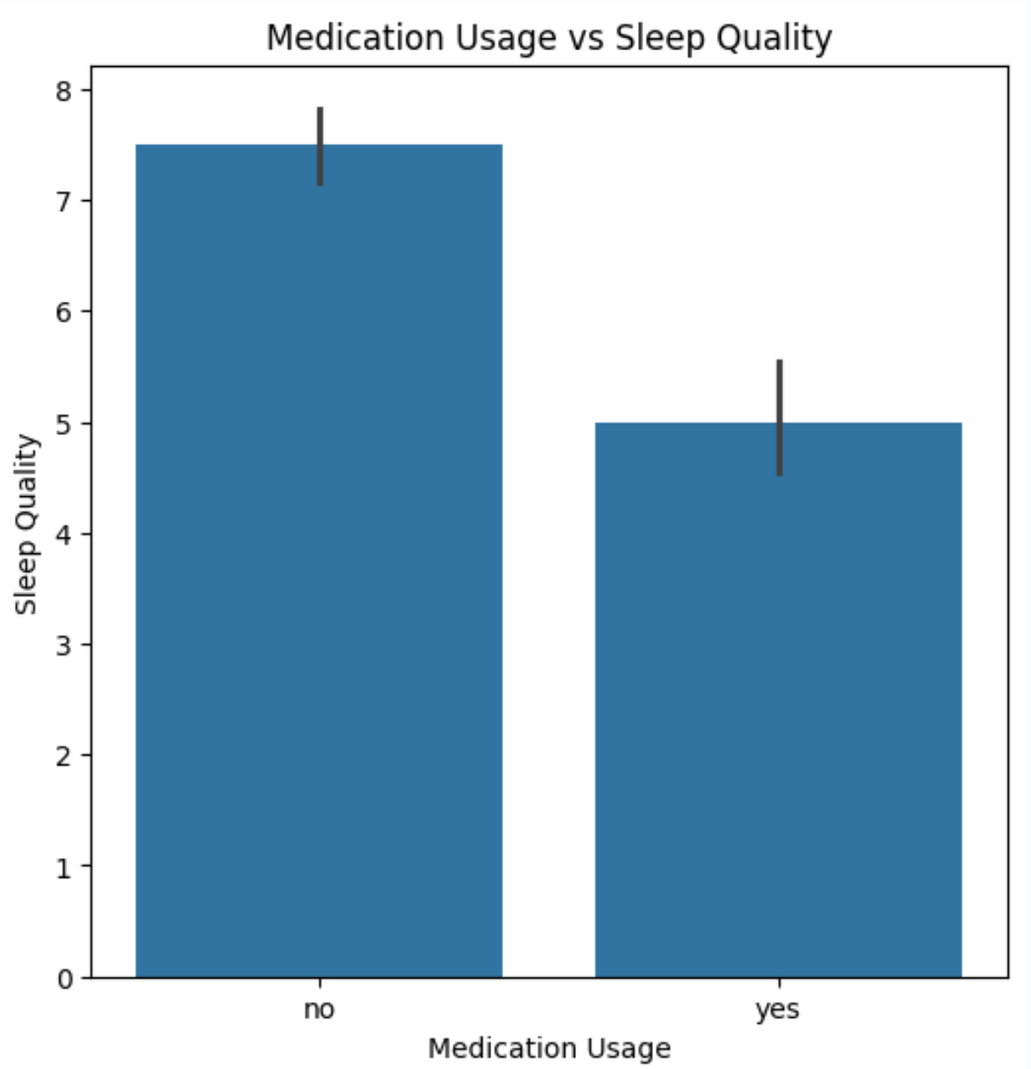
•**Independent T-test** indicates a significant difference in sleep quality between the two groups.

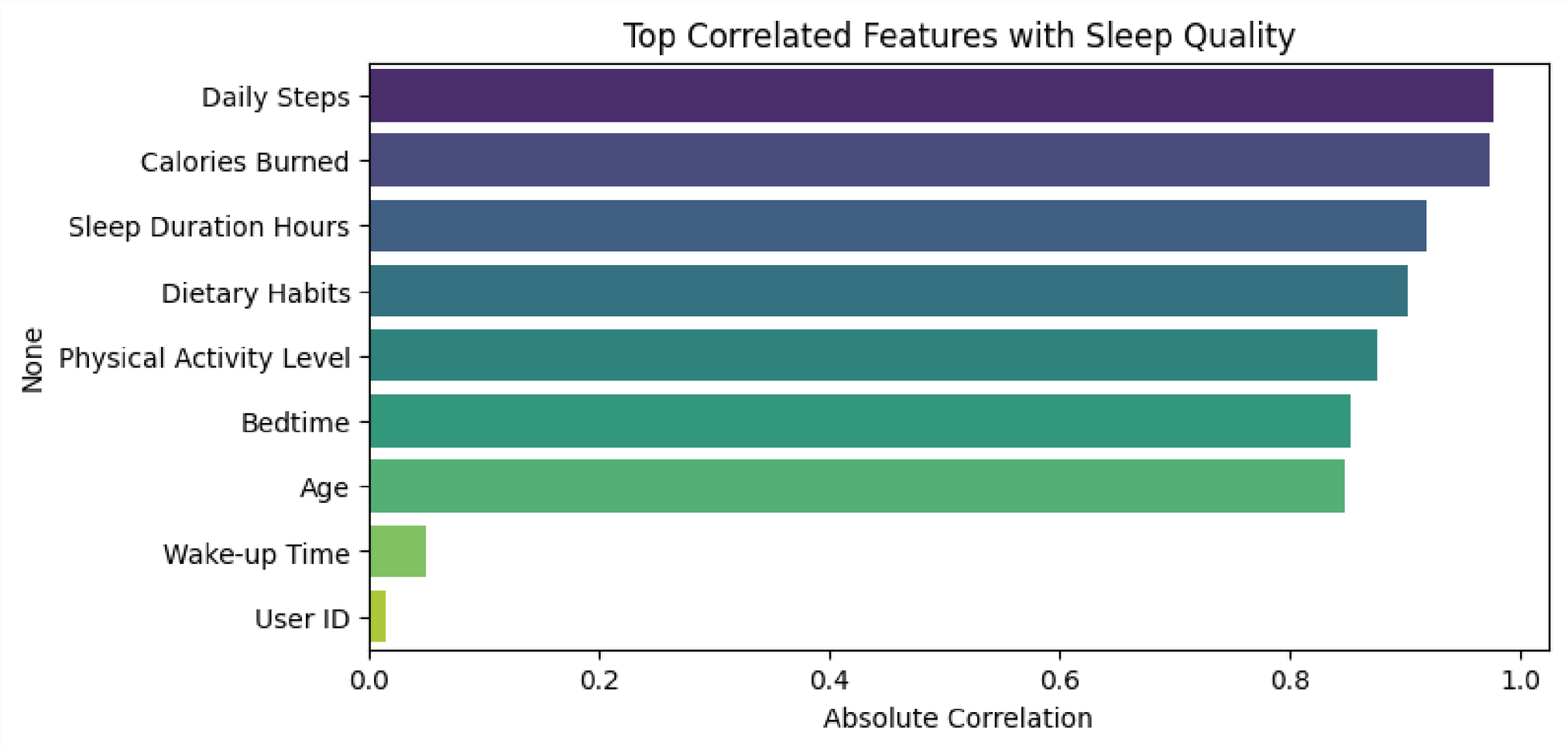
•**t = -8.08**, **p = 0.0000** → medication use is associated with **lower sleep quality**.

Conclusion:

•Taking medication has a **notable impact on reducing sleep quality**.

•This relationship is **statistically strong** and meaningful.




Top Correlated Features with Sleep Quality
Daily Steps
Calories Burned
Sleep Duration Hours
Dietary Habits
Physical Activity Level
Bedtime
Age
Wake-up Time
User ID
None
0.0
0.2
0.4
0.6
0.8
1.0
Absolute Correlation

Impact of Sleep Disorder on Sleep Quality

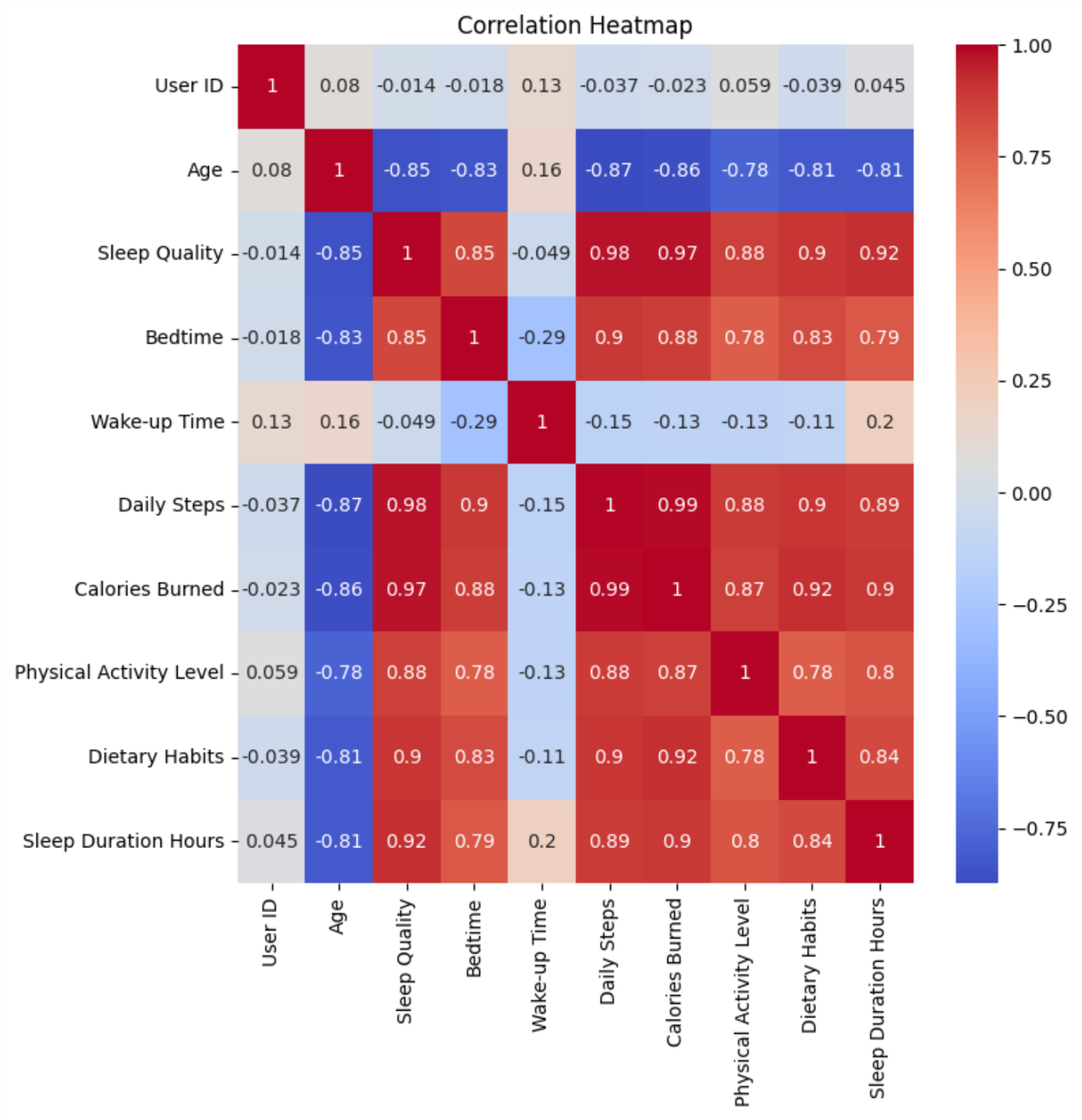
```
df_numeric = df.select_dtypes(include=['number'])
corr = df_numeric.corr()
print("\n=== Correlation with Sleep Quality ===")
print(corr['Sleep Quality'].sort_values(ascending=False))

plt.figure(figsize=(8,8))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Feature	Correlation with Sleep Quality
Sleep Quality	1.0000
Daily Steps	0.9763
Calories Burned	0.9736
Sleep Duration Hours	0.9189
Dietary Habits	0.9025
Physical Activity Level	0.8752
Bedtime	0.8533
User ID	-0.0142
Wake-up Time	-0.0485
Age	-0.8484

Impact of Sleep Disorder on Sleep Quality

Feature	Correlation with Sleep Quality
Sleep Quality	1.0000
Daily Steps	0.9763
Calories Burned	0.9736
Sleep Duration Hours	0.9189
Dietary Habits	0.9025
Physical Activity Level	0.8752
Bedtime	0.8533
User ID	-0.0142
Wake-up Time	-0.0485
Age	-0.8484



Factors Affecting Sleep Quality

1.Daily Steps → 0.976 → more daily steps = better sleep quality.

2.Calories Burned → 0.974 → more daytime activity = improved sleep.

3.Sleep Duration Hours → 0.919 → longer sleep = higher quality.

4.Dietary Habits → 0.902 → healthy diet associated with better sleep.

5.Physical Activity Level → 0.875 → being active improves sleep quality.

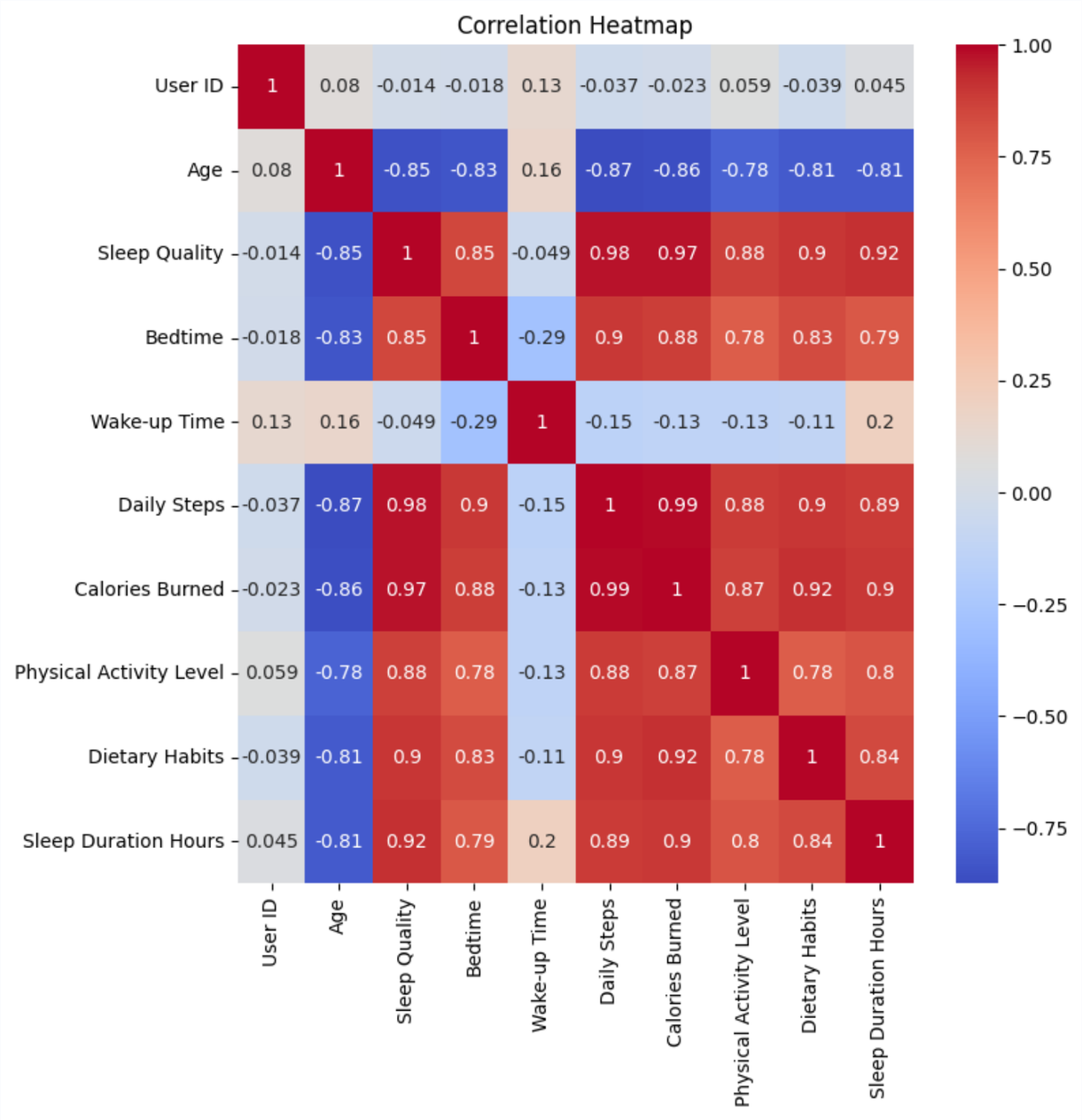
6.Bedtime → 0.853 → regular and earlier sleep linked to better quality.

Strong Negative Correlation:

•**Age** → -0.848 → sleep quality decreases with increasing age.

Low or No Correlation:

•**Wake-up Time** and **User ID** → almost no impact on sleep quality.



Summary:

- Sleep quality is mainly influenced by physical activity, sleep duration and timing, and dietary habits.
- Age plays a significant role in reducing sleep quality.
- These findings can serve as a lifestyle guide for improving sleep health.

Thank You